

Physics, 2017 EC, natural science, entrance exam, round 1

Grade 12 · Physics · 2017 E.C.

Time allowed: 120 minutes

Questions: 45

Total marks: 45

Name _____ Grade _____ ID No. _____ Date _____

1. Which one of the following statements is correct about the current-voltage (I-V) characteristics of a typical semiconductor diode? (1)
 - A. The region of the I-V curve that rises very rapidly after the voltage exceeds the cut-in voltage is the forward bias region.
 - B. The region of the I-V curve that increases very rapidly after the bias voltage exceeds the breakdown voltage is in the forward bias region.
 - C. The current due to the minority charge carriers is represented in the forward bias region of the I-V curve.
 - D. The current due to the majority charge carriers is represented in the reverse bias region of the I-V curve.
2. A force is used to push a box of mass, m along the slope of an inclined plane with constant velocity. If the coefficient of friction between the box and the inclined plane be μ , the actual mechanical advantage (AMA) of the inclined plane is (1)
 - A. $1/(\cos \theta + \mu \sin \theta)$
 - B. $1/(\sin \theta + \mu \cos \theta)$
 - C. $(\cos \theta + \mu \sin \theta)$
 - D. $(\sin \theta + \mu \cos \theta)$
3. A circular coil is located in a uniform magnetic field of magnitude 0.50 T directed perpendicular to the plane of the coil. If the area of the coil changes at a uniform rate, in such a way that it increases by $4.0 \times 10^{-3} \text{ m}^2$ in 2 s, what is the magnitude of the induced electromotive force (emf) in the coil? (1)
 - A. $1.0 \times 10^3 \text{ V}$
 - B. $2.5 \times 10^2 \text{ V}$
 - C. $1.0 \times 10^{-3} \text{ V}$
 - D. $4.0 \times 10^{-3} \text{ V}$
4. Vector quantities are represented by (1)
 - A. direction only.
 - B. magnitude and direction.
 - C. dimensionless values.
 - D. magnitude only.
5. According to Newton's first law, an object that is in motion continues its state of motion with (1)
 - A. constant velocity if the resultant force exerted on it is not zero.
 - B. a decreasing speed if the resultant force exerted on it is zero.
 - C. constant velocity if the resultant force exerted on it is zero.
 - D. an increasing speed if the resultant force exerted on it is not zero.
6. Which of the following correctly describes the difference between distance and displacement? (1)
 - A. Distance depends on the initial and final points, while displacement depends on the path followed by a moving body.
 - B. Distance can be described both in magnitude and direction, while displacement is described only by magnitude.
 - C. Distance has either positive or negative values, while displacement has only positive value.
 - D. Distance is always greater than or equal to the magnitude of displacement, while the magnitude of the displacement is always less than or equal to distance.

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7. Which of the following statements is correct about uniformly accelerated motion? (1)
- A. The acceleration is constant.
 - B. The velocity is constant.
 - C. The displacement increases at a uniform rate.
 - D. The speed is constant but the direction is changing.
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8. Two displacement vectors have magnitudes 4 m and 3 m. Which one of the following is NOT the possible value of the magnitude of their resultant vector? (1)
- A. 7 m
 - B. 1 m
 - C. 12 m
 - D. 5 m
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9. A train is moving along a straight line with a constant acceleration of 5 m/s^2 when it passes by a traffic light point. From there on, if the time taken by the train to reach a velocity of 30 m/s is 4 seconds, then what is the distance of the train from the traffic point after 8 seconds? (1)
- A. 160 m
 - B. 80 m
 - C. 320 m
 - D. 240 m
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10. Which one of the following statements is a necessary condition for an object to be in linear equilibrium? (1)
- A. The linear acceleration of the object is non-zero constant.
 - B. The linear speed of the object is constant.
 - C. The linear acceleration of the object is zero.
 - D. The net force acting on the object is non-zero constant.
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11. When an object tends to slide over the surface of another object, tiny bumps at the surface of the two objects knocking and locking together. This causes (1)
- A. frictional force.
 - B. normal force.
 - C. electrostatic force.
 - D. gravitational force.
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12. Suppose a ball is thrown vertically upward and after reaching a maximum height, it returns back to its initial position. Which of the following statement is correct about energy changes and work done on the ball? (1)
- A. On its way down, change in its potential energy is positive.
 - B. On its way down, work done by gravity is positive.
 - C. On its way up, change in its kinetic energy is positive.
 - D. On its way up, work done by gravity is positive.
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13. The mechanical energy of an oscillating object at any point through its path is equal to (1)
- A. the difference between its kinetic and potential energies.
 - B. the sum of its potential and internal energies.
 - C. the sum of its kinetic and internal energies.
 - D. the sum of its kinetic and potential energies.
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14. A tennis ball of mass 0.10 kg traveling horizontally at 40.0 m/s is struck back by a racket. If the ball returns back with a speed of 30.0 m/s in the opposite direction, what is the magnitude of the impulse delivered to the ball by the racket? (1)
- A. 1.0 kg m/s
 - B. 7.0 kg m/s
 - C. 4.0 kg m/s
 - D. 3.0 kg m/s

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- 15.** The flow rate of a fluid is defined as (1)
- A. the volume of the fluid crossing an area per unit time taken.
 - B. the speed with which the fluid is flowing through a pipe.
 - C. the mass of the fluid crossing an area per unit time taken.
 - D. the rate of change of the speed with which fluid is flowing through a pipe.
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- 16.** At normal condition, solids are almost incompressible compared to liquids and gases. The reason is that (1)
- A. their atoms are free to slide about and interact with neighbors.
 - B. their atoms are relatively fixed distance apart and held together.
 - C. their atoms vibrate and are free to move.
 - D. there is much space and little force between atoms.
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- 17.** Which of the following statements is correct about magnets? (1)
- A. The north pole of one magnet repels the south pole of another magnet.
 - B. Like electric charges, magnetic poles exist in a single isolated form.
 - C. Unlike permanent magnets, electromagnets are magnets without poles.
 - D. If you cut a bar magnet into two, each piece has both north and south poles.
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- 18.** A wheel of radius $r = 50$ cm rotates about a fixed perpendicular axis that passes through its center. If the angular speed is $\omega = 200$ rad/s, how much is the tangential speed of a point on the rim of the wheel? (1)
- A. 10,000 m/s
 - B. 800 m/s
 - C. 80,000 m/s
 - D. 100 m/s
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- 19.** Suppose your mass is measured to be 50 kg. If your friend having a mass of 55 kg is at a distance of 2 m apart from you, what is the magnitude of gravitational force between you and your friend? (1)
- A. 9.2×10^{-8} N
 - B. 9.2×10^{-11} N
 - C. 4.6×10^{-8} N
 - D. 4.6×10^{-11} N
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- 20.** A 10 N force is applied to the smaller circular piston of a hydraulic lift of radius 0.1 m to lift a 2500 N load placed on the larger piston. What is the area of the larger piston? (1)
- A. 2.50 m²
 - B. 0.25 m²
 - C. 3.14 m²
 - D. 7.85 m²
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- 21.** An electric field is defined as a region where (1)
- A. a charged particle loses its charge when it is placed in that region.
 - B. a charged particle experiences no electric force when it is placed in that region.
 - C. a neutral particle experiences electric force when it is placed in that region.
 - D. a charged particle experiences an electric force if it is placed in that region.
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- 22.** Which one of the following statements is correct about the speed of sound in different mediums? The speed of sound (1)
- A. in a colder and less dense medium is faster than in hotter and denser material.
 - B. in solid is less than its speed in liquid.
 - C. increases as the temperature and density of a material increases.
 - D. in liquid is less than its speed in gas.

23. What is the magnitude of electric field strength at a distance of 2.0 m from a point charge, $Q = 4.0 \text{ C}$? (1)

A. $9.0 \times 10^9 \text{ N/C}$ B. $72.0 \times 10^9 \text{ N/C}$
 C. $54.0 \times 10^9 \text{ N/C}$ D. $18.0 \times 10^9 \text{ N/C}$

24. A wire that has a resistance of 6Ω is drawn out to make it into a new wire two times as long as the original and half the cross-sectional area of the original. What will be the resistance of the new wire? (1)

A. 12Ω B. 6Ω
 C. 3Ω D. 24Ω

25. A long straight wire carries a current of 5.0 A. What is the magnetic field strength produced by the current at a distance of 2.0 cm from the wire? (1)

A. $5\pi \times 10^{-5} \text{ T}$ B. $20\pi \times 10^{-5} \text{ T}$
 C. $20 \times 10^{-5} \text{ T}$ D. $5 \times 10^{-5} \text{ T}$

26. If two vectors **a** and **b**, with respective magnitudes of a and b , form an angle α between them when they are connected tail to tail, then the expression $ab \cos \alpha$ defines the value of (1)

A. vector product of **a** and **b**. B. projection of **a** on **b**.
 C. resultant of **a** and **b**. D. scalar product of **a** and **b**.

27. Which one of the following statements is correct about real and virtual images? (1)

A. Real images can be displayed on a screen unlike the virtual ones.
 B. Virtual images can be displayed on a screen unlike the real ones.
 C. Virtual images can be magnified but real images cannot.
 D. Real images can be magnified but virtual images cannot.

28. If the critical angle for an optically denser medium to air boundary is 53° , what will be the refractive index of the medium? (1)

A. 1.67 B. 1.33
 C. 1.00 D. 1.25

29. Given the displacement vector $\mathbf{A} = (3\mathbf{i} - 4\mathbf{j}) \text{ m}$, what is the unit vector in the direction of **A**? (1)

A. $-0.6\mathbf{i} + 0.8\mathbf{j}$ B. $0.6\mathbf{i} - 0.8\mathbf{j}$
 C. $0.8\mathbf{i} - 0.6\mathbf{j}$ D. $0.6\mathbf{i} + 0.8\mathbf{j}$

30. Suppose that you are given two forces with $F_1 = 75 \text{ N}$ in the direction of 53° west of north and $F_2 = 100 \text{ N}$ in the direction of 37° east of north. If you add the two forces using analytical and graphical methods and represent using graphical scale of 1 cm for 10 N force, what are the possible length and direction of the arrow line for the resultant vector? (1)

A. 12.5 cm to the North B. 17.3 cm to 45° North of East
 C. 1.25 cm to the North D. 1.73 cm to 45° North of East

31. Which of the following concepts of physics are the primary focus points in mechanical engineering? (1)

A. Geometric optics, Electrostatics, Thermodynamics, and Stress
 B. Dynamics, Thermodynamics, Force, and Stress
 C. Nuclear physics, Geometric optics, Dynamics, and Force
 D. Molecular physics, Dynamics, Nuclear physics, and Force

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- 32.** Which one of the following defense technologies is correctly matched with its working principle? (1)
- A. Missiles radiate electromagnetic signals and examine the echo received to identify the location of the target.
 - B. A radar is a rocket-propelled weapon designed to deliver an explosive weapon with great accuracy at high speed.
 - C. Infrared devices use heat emission to identify objects that cannot be detected using available light sources.
 - D. Drones radiate electromagnetic signals and examine the echo received to identify the location of the target.
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- 33.** Which one of the following phenomena occurs during the change of phases of matter? During a phase change from (1)
- A. gas to liquid, heat energy is absorbed by a substance while its temperature remains constant.
 - B. solid to liquid, heat energy is released from a substance while its temperature remains constant.
 - C. liquid to solid, heat energy is released from a substance while its temperature changes uniformly.
 - D. liquid to gas, heat energy is absorbed by a substance while its temperature remains constant.
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- 34.** Assume that a ball is thrown horizontally from a building of height h above the ground with initial speed v_0 . Which one of the following statements is correct about the motion of the object? (1)
- A. Its horizontal component acceleration is zero.
 - B. Its vertical component velocity remains constant.
 - C. Its vertical component acceleration is zero.
 - D. Its horizontal component velocity uniformly increases.
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- 35.** A projectile is thrown from level ground with an initial velocity v_0 making an angle of θ above the horizontal. If the maximum height covered by the projectile is H , what is the horizontal range of the projectile in terms of H and θ ? (1)
- A. $H / \tan \theta$
 - B. $H \tan \theta / 4$
 - C. $4H / \tan \theta$
 - D. $H / 4 \tan \theta$
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- 36.** The amount of heat energy required to raise the temperature of a given substance by 1°C is (1)
- A. quantity of heat.
 - B. specific heat capacity.
 - C. latent heat.
 - D. heat capacity.
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- 37.** Which one of the following statements is correct about the properties of waves? (1)
- A. When two waves interfere destructively, the amplitude of the resultant wave is greater than the amplitude of each wave.
 - B. Diffraction of a wave through a narrow gap is more observable than through a wider gap.
 - C. When a wave is reflected, its speed and wavelength are changed.
 - D. When a wave is refracted, its speed and wavelength remain the same.
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- 38.** A square metal plate of coefficient of linear expansion α has a surface area A_0 at a temperature T_0 . If the temperature is raised by an amount ΔT , then the plate surface area is changed to A which is expressed as (1)
- A. $A = A_0(1 + 2\alpha\Delta T)$
 - B. $A = 2A_0(1 + \alpha\Delta T)$
 - C. $A = A_0(1 + \alpha\Delta T)$
 - D. $A = A_0(1 + 3\alpha\Delta T)$

39. An object dropped into a sea sinks to a depth of 100 m. Which of the following is correct about the pressure the object experiences? (1)
- A. The gauge pressure exerted on it is $P_{\text{gauge}} = 1.01 \times 10^5 \text{ Pa}$.
 B. The absolute pressure exerted on it is $P_{\text{absolute}} = 9.30 \times 10^5 \text{ Pa}$.
 C. The absolute pressure exerted on it is $P_{\text{absolute}} = 1.13 \times 10^6 \text{ Pa}$.
 D. The gauge pressure exerted on it is $P_{\text{gauge}} = 1.03 \times 10^7 \text{ Pa}$.
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40. How much heat energy is required to change 5 kg of ice at -20°C to water at 10°C ? (Specific heat capacities of water in liquid and solid phases are $4.2 \text{ kJ/kg} \cdot ^\circ\text{C}$ and $2.1 \text{ kJ/kg} \cdot ^\circ\text{C}$, respectively, and latent heat of fusion of water, $L_f = 336000 \text{ J/kg}$) (1)
- A. $4.20 \times 10^5 \text{ J}$
 B. $1.68 \times 10^6 \text{ J}$
 C. $2.10 \times 10^6 \text{ J}$
 D. $1.89 \times 10^6 \text{ J}$
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41. Which one of the following statements is correct about the fundamental forces in nature and their applications? (1)
- A. The force that keeps the nucleus of an atom from flying apart due to the repulsive force between protons is the weak nuclear force.
 B. The gravitational force makes planets revolve around the Sun and follow their orbital path.
 C. The force that arises in most radioactive processes and in nuclear reactions that generate the Sun's energy is the strong nuclear force.
 D. The electromagnetic force is the force that keeps protons and neutrons bound together in the nucleus of an atom.
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42. A spring stores 120 J of elastic potential energy when it is stretched by 0.5 m. What is the potential energy stored by the spring when it is stretched by 1.5 m? (1)
- A. 480 J
 B. 960 J
 C. 1080 J
 D. 540 J
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43. A crane at port is used to lift a 10,000 kg object to a height of 20 m. If the power developed by the crane is 50,000 W, how long does it take to lift the object? (1)
- A. 400 s
 B. 40 s
 C. 4 s
 D. 2 s
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44. A system of four masses $m_1 = 4 \text{ kg}$, $m_2 = 4 \text{ kg}$, $m_3 = 5 \text{ kg}$, and $m_4 = 3 \text{ kg}$ are located in the xy-plane at the coordinates of (0, 0), (30 cm, 0), (-40 cm, 0), and (0, 20 cm), respectively. What is the coordinate of the center of mass of the system? (1)
- A. (20 cm, 3.75 cm)
 B. (-6.7 cm, 5 cm)
 C. (-5 cm, 10 cm)
 D. (-5 cm, 3.75 cm)
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45. Suppose a particle of mass m is tied to a string and then whirled around in a horizontal circle of radius r about the center O. If the string makes an angle θ to the vertical while rotating, the component of the force responsible for the circular motion of the mass providing the centripetal force is (1)
- A. $T \sin \theta$
 B. $T \cos \theta$
 C. $mg \sin \theta$
 D. $mg \cos \theta$

— END OF EXAMINATION —